RAJALAKSHMI ENGINEERING COLLEGE
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AD23532 – PRINCIPLES OF DATA SCIENCE

EXP 1: Data Preprocessing and Cleaning – Telco Customer Churn

Question:
A telecommunications company is experiencing customer churn. Preprocess the Telco Customer Churn dataset by handling missing values, removing duplicates, encoding categorical features, standardizing numerical attributes, and visualizing important relationships.

AIM:
To perform data preprocessing and cleaning on the Telco Customer Churn dataset by removing duplicates, handling missing values, encoding categorical variables, standardizing numerical features, and visualizing data distributions and correlations.

ALGORITHM:

1. Import the required Python libraries: Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
2. Load the dataset into a Pandas DataFrame and display the first 10 records, last 10 records, shape, and column names.
3. Explore the dataset using info(), describe(), and check missing values and duplicate records.
4. Remove duplicate records and replace missing numerical values with the median and categorical values with the mode.
5. Encode categorical columns using Label Encoding and One-Hot Encoding.
6. Standardize Tenure, MonthlyCharges, and TotalCharges using StandardScaler.
7. Plot the distributions of MonthlyCharges and Tenure, and create a count plot of Contract versus Churn.
8. Draw a box plot of MonthlyCharges grouped by Churn and display the correlation heatmap of numerical features.

PROGRAM & OUTPUT:

```
# Name: Shyam M (241801267)
# Section 1: Import Required Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
# Name: Shyam M (241801267)
# Section 2: Load and Display Dataset

df = pd.read_csv("/WA_Fn-UseC_-Telco-Customer-Churn.csv")

print("First 10 records:\n", df.head(10))
print("Last 10 records:\n", df.tail(10))
print("Shape:", df.shape)
print("Column names:", df.columns.tolist())
```

```
First 10 records:
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes
5  9305-CDSKC  Female              0      No         No       8          Yes
6  1452-KIOVK    Male              0      No        Yes      22          Yes
7  6713-OKOMC  Female              0      No         No      10           No
8  7892-POOKP  Female              0     Yes         No      28          Yes
9  6388-TABGU    Male              0      No        Yes      62          Yes

     MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1               No             DSL            Yes  ...              Yes
2               No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4               No     Fiber optic             No  ...               No
5              Yes     Fiber optic             No  ...              Yes
6              Yes     Fiber optic             No  ...               No
7  No phone service             DSL            Yes  ...               No
8              Yes     Fiber optic             No  ...              Yes
9               No             DSL            Yes  ...               No

   TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0           No          No              No  Month-to-month              Yes
1           No          No              No        One year               No
2           No          No              No  Month-to-month              Yes
3          Yes          No              No        One year               No
4           No          No              No  Month-to-month              Yes
5           No         Yes             Yes  Month-to-month              Yes
6           No         Yes              No  Month-to-month              Yes
7           No          No              No  Month-to-month               No
8          Yes         Yes             Yes  Month-to-month              Yes
```

```
0       Yes        Yes             Yes  Month-to-month              Yes
9        No         No              No        One year               No
...
              PaymentMethod  MonthlyCharges  TotalCharges Churn
0          Electronic check           29.85         29.85    No
1              Mailed check           56.95        1889.5    No
2              Mailed check           53.85        108.15   Yes
3  Bank transfer (automatic)           42.30       1840.75    No
4          Electronic check           70.70        151.65   Yes
5          Electronic check           99.65         820.5   Yes
6   Credit card (automatic)           89.10        1949.4    No
7              Mailed check           29.75          301.9    No
8          Electronic check          104.80       3046.05   Yes
9  Bank transfer (automatic)           56.15       3487.95    No

[10 rows x 21 columns]
Last 10 records:
      customerID  gender  SeniorCitizen Partner Dependents  tenure  \
7033  9767-FFLEM    Male              0      No         No      38
7034  0639-TSIQW  Female              0      No         No      67
7035  8456-QDAVC    Male              0      No         No      19
7036  7750-EYXWZ  Female              0      No         No      12
7037  2569-WGERO  Female              0      No         No      72
7038  6840-RESVB    Male              0     Yes        Yes      24
7039  2234-XADUH  Female              0     Yes        Yes      72
7040  4801-JZAZL  Female              0     Yes        Yes      11
7041  8361-LTMKD    Male              1     Yes         No       4
7042  3186-AJIEK    Male              0      No         No      66

     PhoneService     MultipleLines InternetService       OnlineSecurity  ...
7033          Yes               No     Fiber optic                   No  ...
7034          Yes              Yes     Fiber optic                  Yes  ...
7035          Yes               No     Fiber optic                   No  ...
7036           No  No phone service             DSL                   No  ...
7037          Yes               No              No  No internet service  ...
7038          Yes              Yes             DSL                  Yes  ...
7039          Yes              Yes     Fiber optic                   No  ...
7040           No  No phone service             DSL                  Yes  ...
7041          Yes              Yes     Fiber optic                   No  ...
7042          Yes               No     Fiber optic                  Yes  ...

          DeviceProtection          TechSupport          StreamingTV  \
7033                    No                   No                   No
7034                   Yes                   No                  Yes
7035                    No                   No                  Yes
7036                   Yes                  Yes                  Yes
7037  No internet service  No internet service  No internet service
7038                   Yes                  Yes                  Yes
7039                   Yes                   No                  Yes
7040                    No                   No                   No
```

```
7033                  No                   No                  Yes
7034                 Yes                  Yes                  Yes
7035  No internet service  No internet service  No internet service
7036                 Yes                  Yes                  Yes
7037                 Yes                   No                  Yes
7038                  No                   No                   No
7039                 Yes                  Yes                  Yes
7040                  No                   No                   No
7041                  No                   No                   No
7042                 Yes                  Yes                  Yes

      StreamingMovies        Contract PaperlessBilling  \
7033              No  Month-to-month              Yes
7034              No  Month-to-month              Yes
7035              No  Month-to-month              Yes
7036             Yes        One year               No
7037  No internet service        Two year              Yes
7038             Yes        One year              Yes
7039             Yes        One year              Yes
7040              No  Month-to-month              Yes
7041              No  Month-to-month              Yes
7042             Yes        Two year              Yes

              PaymentMethod  MonthlyCharges  TotalCharges Churn
7033   Credit card (automatic)           69.50       2625.25    No
7034   Credit card (automatic)          102.95       6886.25   Yes
7035  Bank transfer (automatic)           78.70       2481.1    No
7036          Electronic check           60.65        743.3    No
7037  Bank transfer (automatic)           21.15       1419.4    No
7038              Mailed check           84.80       1990.5    No
7039   Credit card (automatic)          103.20       7362.9    No
7040          Electronic check           29.60        346.45   No
7041              Mailed check           74.40        306.6    Yes
7042  Bank transfer (automatic)          105.65       6844.5    No

[10 rows x 21 columns]
Shape: (7043, 21)
Column names: ['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'St
```

```
# Name: Shyam M (241801267)
# Section 3: Explore Dataset

df.info()
print(df.describe(include="all"))
print("Missing values:\n", df.isnull().sum())
print("Duplicate records:", df.duplicated().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
        customerID gender  SeniorCitizen Partner Dependents       tenure  \
count         7043   7043    7043.000000    7043        7043  7043.000000
unique        7043      2            NaN       2           2          NaN
top     3186-AJIEK   Male            NaN      No          No          NaN
freq             1   3555            NaN    3641        4933          NaN
mean           NaN    NaN       0.162147     NaN         NaN    32.371149
std            NaN    NaN       0.368612     NaN         NaN    24.559481
min            NaN    NaN       0.000000     NaN         NaN     0.000000
25%            NaN    NaN       0.000000     NaN         NaN     9.000000
50%            NaN    NaN       0.000000     NaN         NaN    29.000000
75%            NaN    NaN       0.000000     NaN         NaN    55.000000
max            NaN    NaN       1.000000     NaN         NaN    72.000000
```

```
        PhoneService MultipleLines InternetService OnlineSecurity  ... \
count           7043          7043            7043           7043  ...
unique             2             3               3              3  ...
top              Yes            No     Fiber optic             No  ...
freq            6361          3390            3096           3498  ...
mean             NaN           NaN             NaN            NaN  ...
std              NaN           NaN             NaN            NaN  ...
min              NaN           NaN             NaN            NaN  ...
25%              NaN           NaN             NaN            NaN  ...
50%              NaN           NaN             NaN            NaN  ...
75%              NaN           NaN             NaN            NaN  ...
max              NaN           NaN             NaN            NaN  ...

       DeviceProtection TechSupport StreamingTV StreamingMovies  \
count              7043        7043        7043            7043
unique                3           3           3               3
top                  No          No          No              No
freq               3095        3473        2810            2785
mean                NaN         NaN         NaN             NaN
std                 NaN         NaN         NaN             NaN
min                 NaN         NaN         NaN             NaN
25%                 NaN         NaN         NaN             NaN
50%                 NaN         NaN         NaN             NaN
75%                 NaN         NaN         NaN             NaN
max                 NaN         NaN         NaN             NaN

             Contract PaperlessBilling     PaymentMethod MonthlyCharges  \
count            7043             7043              7043    7043.000000
unique              3                2                 4            NaN
top     Month-to-month              Yes  Electronic check            NaN
freq             3875             4171              2365            NaN
mean              NaN              NaN               NaN      64.761692
std               NaN              NaN               NaN      30.090047
min               NaN              NaN               NaN      18.250000
25%               NaN              NaN               NaN      35.500000
50%               NaN              NaN               NaN      70.350000
75%               NaN              NaN               NaN      89.850000
max               NaN              NaN               NaN     118.750000
```

```
       TotalCharges Churn
count          7043  7043
unique         6531     2
top                    No
freq             11  5174
mean            NaN   NaN
std             NaN   NaN
min             NaN   NaN
25%             NaN   NaN
50%             NaN   NaN
75%             NaN   NaN
max             NaN   NaN

[11 rows x 21 columns]
Missing values:
 customerID          0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges         0
Churn                0
dtype: int64
Duplicate records: 0
```

```
# Name: Shyam M (241801267)
# Section 4: Clean Dataset

df.replace(r'^\s*$', np.nan, regex=True, inplace=True)

if "TotalCharges" in df.columns:
    df["TotalCharges"] = pd.to_numeric(
        df["TotalCharges"], errors="coerce"
    )

df.drop_duplicates(inplace=True)

for col in df.columns:
    if df[col].isnull().any():
        if pd.api.types.is_numeric_dtype(df[col]):
            df[col] = df[col].fillna(df[col].median())
        else:
            df[col] = df[col].fillna(df[col].mode()[0])

print("Missing values after cleaning:\n", df.isnull().sum())
```

```
Missing values after cleaning:
 customerID          0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges         0
Churn                0
dtype: int64
```

```
# Name: Shyam M (241801267)
# Section 5: Encode Categorical Columns

binary_cols = [
    c for c in [
        "gender", "Partner", "Dependents", "Churn",
        "PhoneService", "PaperlessBilling"
    ] if c in df.columns
]

le = LabelEncoder()

for col in binary_cols:
    df[col] = le.fit_transform(df[col].astype(str))

multi_cols = [
    c for c in [
        "Contract", "PaymentMethod", "InternetService"
    ] if c in df.columns
]

df = pd.get_dummies(df, columns=multi_cols, dtype=int)
```

```
# Name: Shyam M (241801267)
# Section 6: Standardize Numerical Features

num_cols = [
    c for c in ["tenure", "MonthlyCharges", "TotalCharges"]
    if c in df.columns
]

scaler = StandardScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])

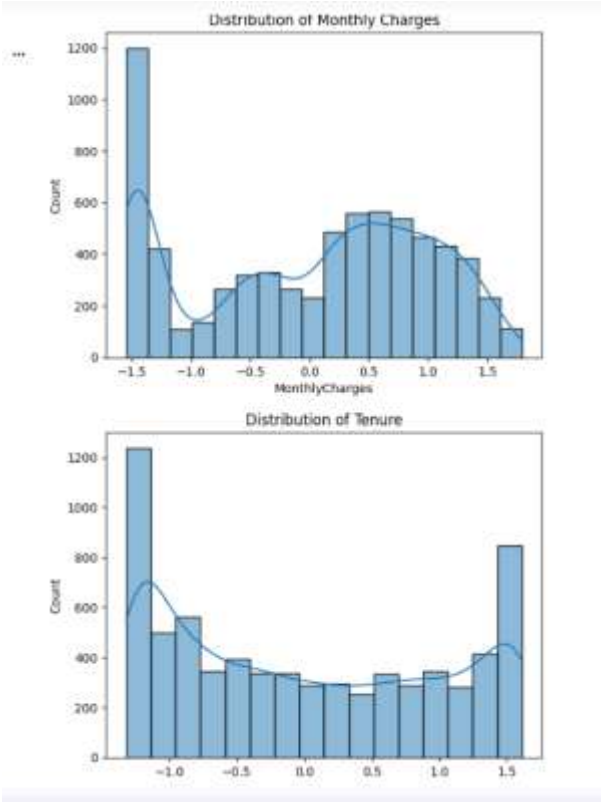
print("Standardized numerical features:\n", df[num_cols].head())
```

```
Standardized numerical features:
      tenure  MonthlyCharges  TotalCharges
0 -1.277445       -1.160323     -0.994242
1  0.066327       -0.259629     -0.173244
2 -1.236724       -0.362660     -0.959674
3  0.514251       -0.746535     -0.194766
4 -1.236724        0.197365     -0.940470
```

```
# Name: Shyam M (241801267)
# Section 7: Plot Data Distributions

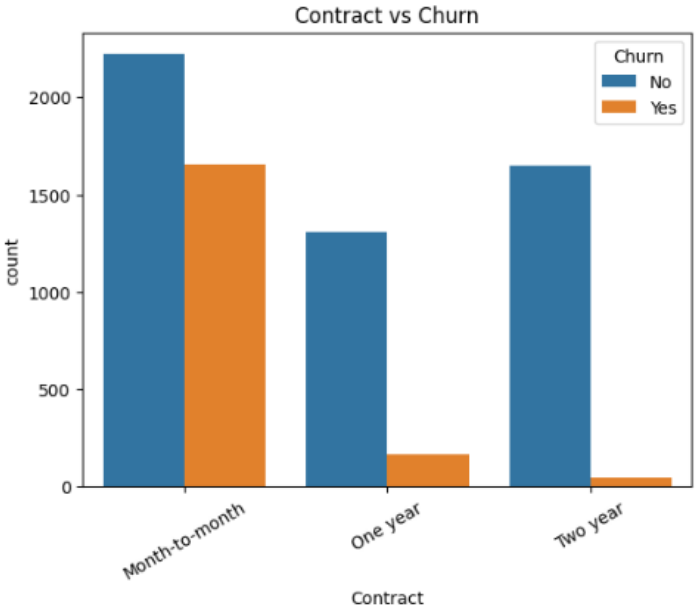
sns.histplot(df["MonthlyCharges"], kde=True)
plt.title("Distribution of Monthly Charges")
plt.show()

sns.histplot(df["tenure"], kde=True)
plt.title("Distribution of Tenure")
plt.show()
```



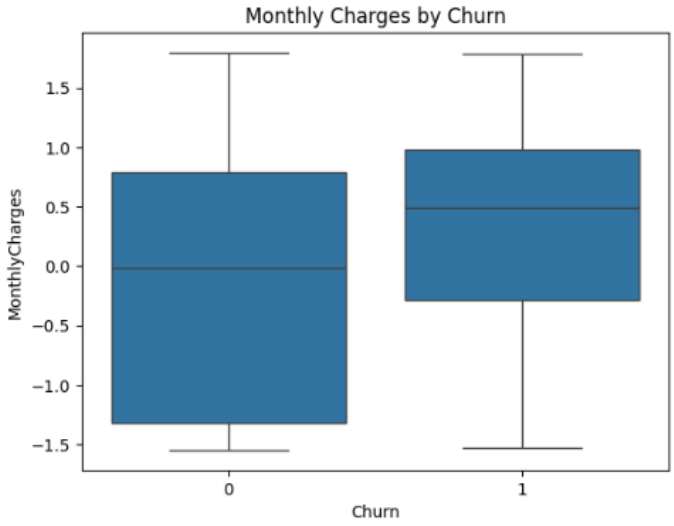
```
# Name: Shyam M (241801267)
# Section 8: Count Plot of Contract vs Churn

sns.countplot(data=df_plot, x="Contract", hue="Churn")
plt.title("Contract vs Churn")
plt.xticks(rotation=30)
plt.show()
```




```
# Name: Shyam M (241801267)
# Section 9: Box Plot of Monthly Charges by Churn

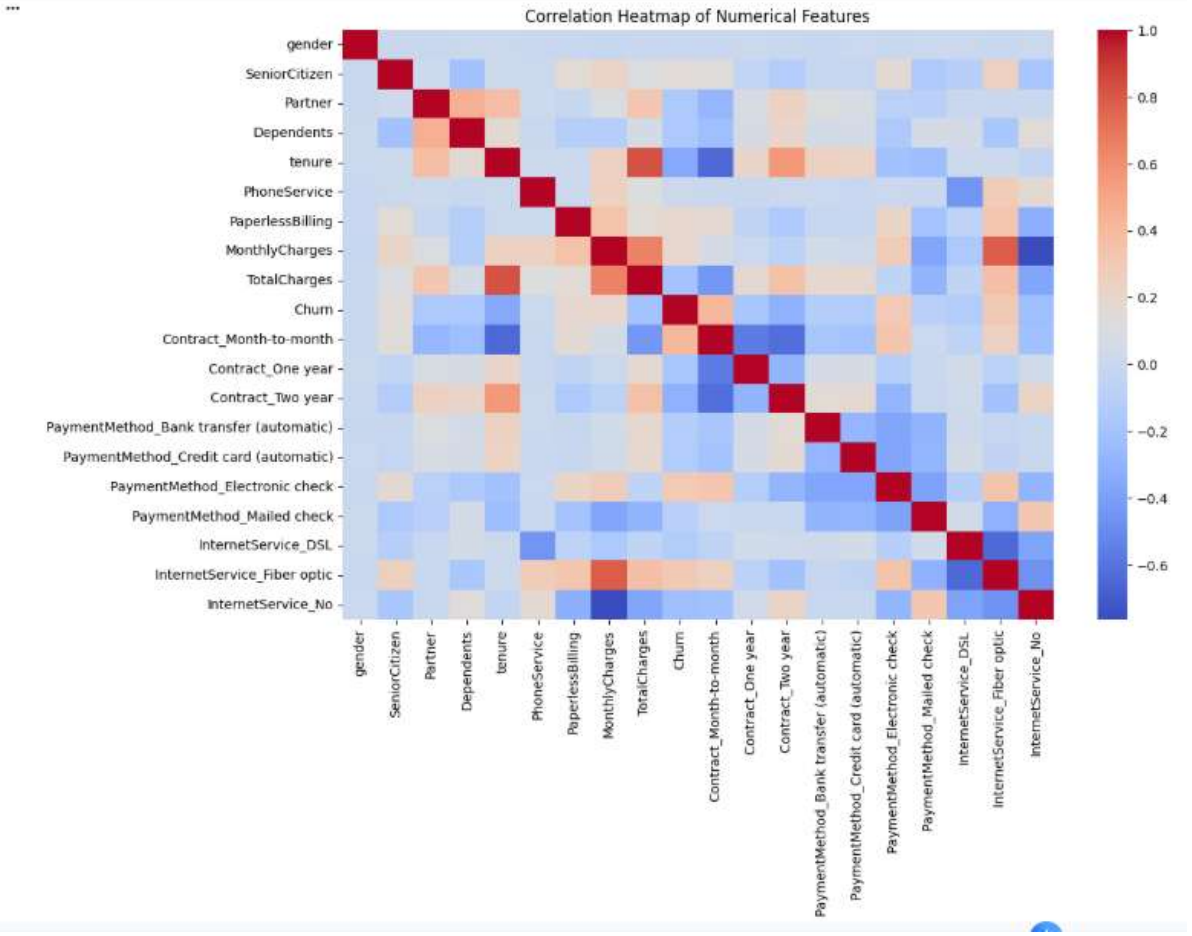
sns.boxplot(data=df, x="Churn", y="MonthlyCharges")
plt.title("Monthly Charges by Churn")
plt.show()
```



```
# Name: Shyam M (241801267)
# Section 10: Correlation Heatmap

numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(12, 8))
sns.heatmap(numeric_df.corr(), cmap="coolwarm", annot=False)
plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```



```
# Name: Shyam M (241801267)
# Section 11: Display Final Dataset

print("Preprocessing completed successfully.")
print(df.head())
print("Final dataset shape:", df.shape)
```

```
Preprocessing completed successfully.
   customerID  gender  SeniorCitizen  Partner  Dependents    tenure  \
0  7590-VHVEG       0              0        1           0 -1.277445
1  5575-GNVDE       1              0        0           0  0.066327
2  3668-QPYBK       1              0        0           0 -1.236724
3  7795-CFOCW       1              0        0           0  0.514251
4  9237-HQITU       0              0        0           0 -1.236724

   PhoneService     MultipleLines OnlineSecurity OnlineBackup  ...  \
0             0  No phone service             No          Yes  ...
1             1                No            Yes           No  ...
2             1                No            Yes          Yes  ...
3             0  No phone service            Yes           No  ...
4             1                No             No           No  ...

  Contract_Month-to-month Contract_One year Contract_Two year  \
0                       1                0                 0
1                       0                1                 0
2                       1                0                 0
3                       0                1                 0
4                       1                0                 0

  PaymentMethod_Bank transfer (automatic)  \
0                                       0
1                                       0
2                                       0
3                                       1
4                                       0

   PaymentMethod_Credit card (automatic)  PaymentMethod_Electronic check  \
0                                      0                               1
1                                      0                               0
2                                      0                               0
3                                      0                               0
4                                      0                               1

   PaymentMethod_Mailed check  InternetService_DSL  \
0                           0                    1
1                           1                    1
2                           1                    1
3                           0                    1
4                           0                    0

   InternetService_Fiber optic  InternetService_No
0                            0                   0
1                            0                   0
2                            0                   0
3                            0                   0
4                            1                   0

[5 rows x 28 columns]
Final dataset shape: (7043, 28)
```

RESULT:

Thus, the Telco Customer Churn dataset was preprocessed successfully by removing duplicate records, handling missing values, encoding categorical variables, and standardizing numerical features. The distributions of Monthly Charges and Tenure, Contract versus Churn count plot, box plot of Monthly Charges grouped by Churn, and correlation heatmap were generated to visualize and understand the dataset.